MODULE 1 · WINDSHEAR

Lesson 01: Windshear: Definition, Sources and Effects

Lesson 1 of 5 · about 7 min

LEARNING OBJECTIVES

- Define windshear and microburst.
- Identify the weather situations that produce low-level windshear.
- Explain the effect of windshear on aircraft performance.



Microbursts from convective clouds are the most dangerous windshear.

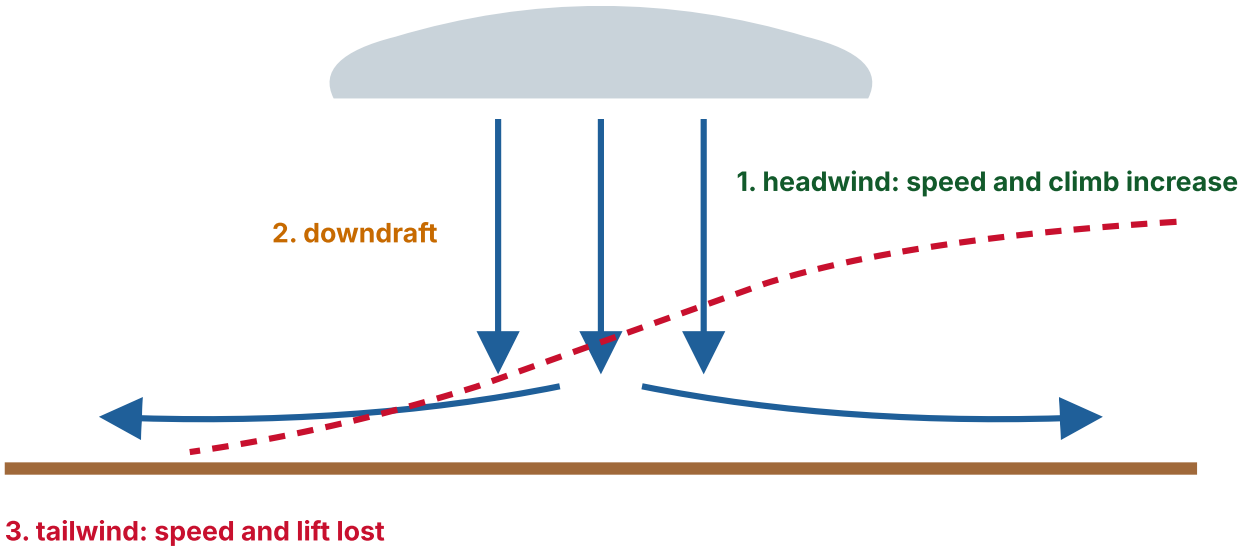
Definition

Windshear is a change of wind speed and/or direction over a short distance, horizontally or vertically. **Low-level windshear** — below about 1,600 ft above ground — is the most dangerous, because it occurs during take-off and approach, when the aircraft is slow, close to the ground and has little energy margin (ICAO Doc 9817).

Sources

- **Thunderstorms and convective clouds** — downbursts and microbursts, gust fronts and outflow, which can extend well away from the cell.
- **Frontal systems** — especially fast-moving cold fronts with strong temperature contrasts.
- **Low-level jet streams and temperature inversions** — wind changing sharply in speed or direction across the inversion layer.
- **Terrain and obstacles** — strong winds over hills, buildings or hangars near the runway, and sea breezes.

The microburst



A microburst first gives a performance-increasing headwind, then a downdraft and a performance-decreasing tailwind. Avoid it; if caught, apply the windshear escape manoeuvre.

A **microburst** is a small, intense downdraft — typically less than about 4 km across — that spreads out on reaching the ground. It usually lasts only a few minutes. An aircraft entering it first meets an increasing headwind (performance increase), then a strong downdraft, then a tailwind (performance decrease). The pilot's natural reaction to the first phase — reducing thrust and pitch — leaves the aircraft badly placed for the dangerous second and third phases.

Effects

- An increasing headwind or decreasing tailwind raises indicated airspeed and lift — the aircraft tends to rise above the path.
- A decreasing headwind or increasing tailwind reduces indicated airspeed and lift — the aircraft sinks below the path.
- A downdraft directly increases the descent rate.
- The combination can exceed the aircraft's climb performance close to the ground.

KEY TAKEAWAYS

- Windshear: change of wind speed or direction over a short distance; most dangerous below about 1,600 ft.
- Sources: convection (microbursts), fronts, inversions and low-level jets, terrain.
- Microburst: headwind gain, then downdraft, then tailwind loss — do not react to the first phase by reducing energy.

KNOWLEDGE CHECK

Entering a microburst on approach, what is usually encountered first?

A. A tailwind

B. An increasing headwind and a performance increase

C. A strong downdraft

D. No change

Answer: B. The first phase is an increasing headwind that raises airspeed and lift — followed by downdraft and tailwind.

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